Module 7

# Capital Budgeting

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

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# Goals

- How to decide what project(s) to do? The tool is the NPV rule
- But to use NPV, we need a project's cash flows
- Here, combine what we know about accounting and cash flow effects (**Module 2**) with the NPV rule (**Module 6**), to decide whether or not to accept a project
- **Capital budgeting**, the process of analyzing capital investment opportunities ("projects") and deciding which ones to accept

# Roadmap

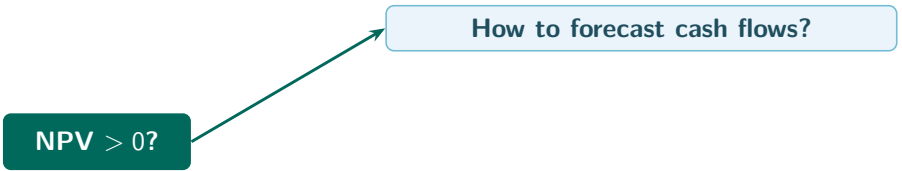
## Should we invest in a project (e.g., a data center)?

**NPV $> 0$?**

Recall from **Module 6**

# Roadmap

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## Should we invest in a project (e.g., a data center)?

**How to forecast cash flows?**

- Similar to **Module 2**, indirect method for cash flow
- But different, cash flow from a project, not a whole firm

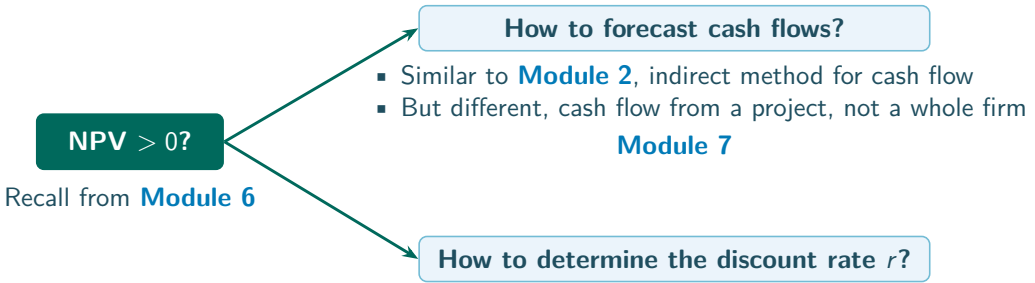
**Module 7**

**NPV $> 0$?**

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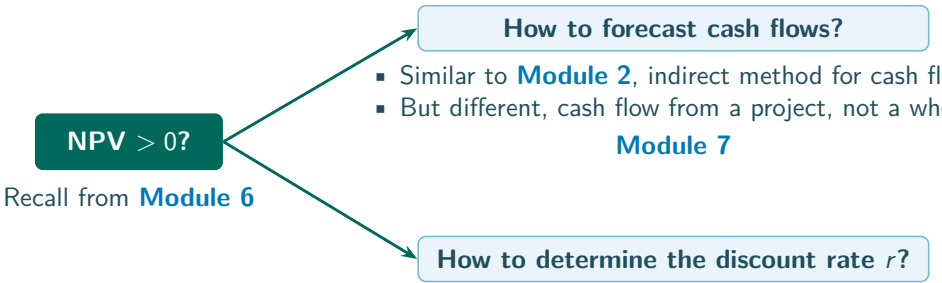
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# Motivating Example, Big Tech's AI Spending




Reuters, October 19, 2026 (5:54), "Big tech risks outspending its AI returns"

## Module 7, Part 0

# Review of the Statement of Cash Flows

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# Statement of Cash Flows

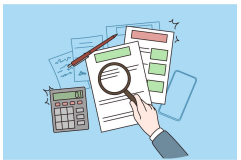
Recall from **Module 2**

| Direct method | Indirect method |
|---|---|
| Lists every cash receipt and payment | Starts from net income and adjusts it |
| More time-consuming to prepare | Faster to prepare |
| Rarely used in practice | **The common choice** |

**Direct** →



**Indirect** →



# Statement of Cash Flows

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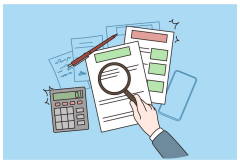
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**Indirect** →



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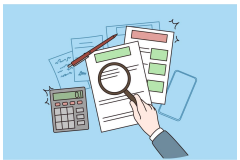
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- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

**Direct** →



**Indirect** →



# How Do Net Income and Cash Flow Differ?

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***Relationship of cash flow (CF) and net income (NI)?***
***When spending on fixed assets, CF = NI − spending on fixed assets***
***When recording depreciation, CF = NI + depreciation***

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Recall from **Module 2**

- The firm issues (sells) $1 billion of new stock to investors

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***When raising money, CF = NI + amount raised***

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- None of them have been sold yet




OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

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***When adding to inventories, CF = NI − change in inventories***

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Recall from **Module 2**

- The firm owed suppliers $30 million for electronic parts
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***When firm pays off suppliers, CF = NI + change in accounts payable***

# Operating Activities

Recall from **Module 2**

**Operating activities**, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
   - NWC = inventories + accounts receivable − accounts payable
   - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

# Why Subtract Change in Net Working Capital (NWC)?

Recall from **Module 2**

***When adding to inventories, CF = NI − change in inventories***

***When firm not yet paid by customers, CF = NI − change in accounts receivable***

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$$CF = NI - \text{change in (inventories} + \text{accounts receivable} - \text{accounts payable)}$$

$$CF = NI - \text{change in net working capital (NWC)}$$

# Operating Activities

Recall from **Module 2**

| Cash flow from operating activities | |
|---|---|
| Net income | 3,160 |
| Depreciation | 3,000 |
| Change in net working capital | −1,000 |
| Cash flow from operating activities | 7,160 |

$$3{,}160 + 3{,}000 - (-1{,}000) = 7{,}160$$

# Investing Activities

Recall from **Module 2**

**Investing activities**, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold
   - **Capital**, a firm's fixed assets
   - **Capital expenditure**, the amount spent on buying capital during the period
   - Cash flow is *negative* if firms buy capital

# Investing Activities

Recall from **Module 2**

| Cash flow from investing activities | |
|---|---|
| Cash flow from capital expenditure | −3,000 |
| Cash flow from sales of fixed assets | 200 |
| Cash flow from investing activities | −2,800 |

$$-3{,}000 + 200 = -2{,}800$$

# Financing Activities

Recall from **Module 2**

**Financing activities**, cash raised from lenders and shareholders, and cash returned to them

# Recap of Part 0

Recall from **Module 2**

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
  - **Operating activities**, cash flow = Net income + Depreciation − Change in NWC
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

## Module 7, Part 1

# Project Cash Flow

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# Firm Cash Flow vs. Project Cash Flow

- **Module 2** looked at the whole firm, here we look at one project on its own
- **Operating activities**, the same idea, project net income plus depreciation minus the change in NWC
- **Investing activities**, the project's capital investments, buying its assets at the start and selling them at the end
- **Financing activities**, none, we do not consider financing activities when evaluating a project on its own

# Identify Project Cash Flow

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$$
\begin{aligned}
\text{Cash flow from a project} &= \text{Project net income} \\
&+ \text{Depreciation} \\
&+ \text{Cash flow from changes in working capital} \\
&+ \text{Cash flow from capital investments}
\end{aligned}
$$

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| |
|---|
| Revenue from the project |
| − Cash expenses for the project |
| − Depreciation for the project's capital investment |
| Project EBIT |
| − Taxes (tax rate × project EBIT) |
| Project net income |

# Component 2, Depreciation

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- Depreciation is subtracted on the project's income statement, but it is not a cash flow, the cash went out when the project's capital investment was made
- To get cash flow from the project, add it back, as in **Module 2**

# Example, Project Net Income and Depreciation

- A project requires an investment of $2,000. The project's revenues minus its cash expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%

# Example, Project Net Income and Depreciation

- A project requires an investment of $2,000. The project's revenues minus its cash expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%
- Calculate the project's net income plus depreciation for year 1

# Example, Project Net Income and Depreciation

- Solution,

| Item | $ |
|---|---|
| Revenues − Cash expenses | 1,100 |
| − Depreciation | 1,000 |
| Project EBIT | 100 |
| − Taxes | 21 |
| Project net income | 79 |
| + Depreciation | 1,000 |
| Project net income + Depreciation | 1,079 |

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- Solution,

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- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$

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- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$
- Project net income + Depreciation $= 79 + 1{,}000 = \$1{,}079$

# Component 3, Cash Flow from Changes in Working Capital

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# Component 3, Cash Flow from Changes in Working Capital

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- A project has its own NWC, e.g., an inventory of raw materials before production starts, and accounts receivable rising as sales start

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- A project has its own NWC, e.g., an inventory of raw materials before production starts, and accounts receivable rising as sales start
- As in Module 2, an increase in the project's NWC is a negative cash flow, and a decrease is a positive cash flow,

  Cash flow from changes in working capital = − Change in the project's NWC

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|---|---|---|---|---|
| The project's NWC | 100 | 100 | 100 | 0 |
| Change in NWC | $+100$ | 0 | 0 | $-100$ |
| Cash flow from changes in working capital | $-100$ | 0 | 0 | $+100$ |

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- Cash goes out up front, as the project stocks up and waits to be paid, and comes back at the end, when the stock is sold and the bills are collected

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- Cash goes out up front, as the project stocks up and waits to be paid, and comes back at the end, when the stock is sold and the bills are collected
- Undiscounted the two cancel, but the \$100 goes out at $t = 0$ and returns only at $t = 3$, so tying up the cash still costs the time value of money

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- If the project requires spending money on its own capital investment to get started (e.g., purchasing a building), that is a negative cash flow
- Usually depreciated over time, in this course always "straight-line" depreciation,

$$\text{Annual depreciation} = \frac{\text{Investment} - \text{Salvage value}}{\text{Expected life}}$$

# Recap of Part 1

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- The project's capital investments are negative cash flows when the assets are bought, depreciated straight-line over the project's life

## Module 7, Part 2

# Putting It All Together

# Example, Putting It All Together

- A company is considering building a new data center which costs \$1 billion (\$1,000 million) and will last 5 years. It will be depreciated straight-line over 5 years with a salvage value of \$200 million. The data center can also be sold at $t = 5$ for \$200 million

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- The corporate tax rate is 21%, and the discount rate is 8%
- Should the company build the data center or not?

# Example, Putting It All Together

- Solution ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Revenues − Cash expenses | | 200 | 200 | 200 | 200 | 200 |
| Depreciation | | 160 | 160 | 160 | 160 | 160 |
| EBIT | | 40 | 40 | 40 | 40 | 40 |
| Net income | | 31.6 | 31.6 | 31.6 | 31.6 | 31.6 |
| + Depreciation | | 160 | 160 | 160 | 160 | 160 |
| Operating cash flows | | 191.6 | 191.6 | 191.6 | 191.6 | 191.6 |
| Capital investment | −1,000 | | | | | |
| Sale of the data center | | | | | | 200 |
| Working capital | −100 | | | | | 100 |
| **Total cash flows** | −1,100 | 191.6 | 191.6 | 191.6 | 191.6 | 491.6 |
| **NPV** | −130.82 | | | | | |

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  - **Operating cash flow before accounting for $\Delta$NWC**, in years 1–5 the rental income adds 200 to revenues minus cash expenses. Depreciation rises by $(1{,}000 - 200)/5 = 160$ per year, a net 40 higher EBIT. After taxes and adding back depreciation, operating cash flow rises by 191.6 each year

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- Putting it all together, NPV = −$130.82 million, so the company should not build the data center

## Module 7, Part 3

# The Principles behind Project Cash Flows

# General Principles for Identifying Project Cash Flow

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- **Accounting approach** (the **wrong** approach, it can lead to erroneous decisions), consider accounting income
- In the data center table, we discounted the project's operating cash flows ($191.6 million a year), not its net income ($31.6 million)

# Cash Flow vs. Accounting Income

- A project costs $2,000 and is expected to last 2 years, producing cash income of $1,500 and $500, respectively. The cost can be depreciated at $1,000 per year. Given a 10% required return, compare the NPV using accounting income to the NPV using cash flow

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- This project is clearly a loser!

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  - The effects of interest (and the interest tax shield) enter when we *separately* consider financing decisions
- The data center table has no interest line, even if the company borrows to build it

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- Depreciation is not a cash flow, but it lowers taxable income and so the taxes the project pays, the depreciation tax shield
- Different depreciation schedules, in this course we stick to "straight-line" depreciation, as in the data center, $(1{,}000 - 200)/5 = \$160$ million a year

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$$\text{Incremental cash flow} = \text{Cash flow with the project} - \text{Cash flow without it}$$

- In the data center table, the rental income and the spare parts exist only because of the data center, so they count

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e. Recognize investments in working capital
f. Remember terminal cash flows

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- But the launch will hurt sales of the older console, forecast to create \$800m and \$1,000m in years 1 and 2, only \$500m and \$400m with the new one on the market
- What is the **incremental** cash flow of the new console?






# Rule a, Indirect Effects

- Solution ($ millions),

| | $t = 0$ | $t = 1$ | $t = 2$ |
|---|---|---|---|
| Cash flow from the old console without the new one | | 800 | 1,000 |
| Cash flow from the old console with the new one | | 500 | 400 |
| Cash flow from investment in and sales of the new one | $-200$ | 400 | 800 |
| **Incremental cash flow of the new console** | $-200$ | $400 - 300 = 100$ | $800 - 600 = 200$ |

# Rule b, Sunk Costs

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- Many sunk costs are "related" to a project, but no longer incremental if the current decision does not affect them



# Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
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- Is this project a good idea today?
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- What is the actual cost of building the factory?

# Rule d, Overhead

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- The principle of incremental cash flows says only *changes* in overhead that result from a project count

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- Working capital often keeps changing during the project, and is often recovered at the end
- Common mistakes,
  1. Forgetting about working capital entirely
  2. Forgetting that working capital may change during the life of the project
  3. Forgetting that working capital is often recovered at the end of the project

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- E.g., in the data center example, suppose it could be sold for \$500 million at $t = 5$. It has been depreciated to \$200 million, so the company pays taxes on the gain,

$$(500 - 200) \times 21\% = \$63 \text{ million}$$

# Rule f, Terminal Cash Flows

- At the end, there may be cash flows from shutting down the project, the "terminal" cash flows
  - E.g., selling the machines and equipment of the project, a positive cash flow!
- Keep track of the tax consequences, if capital is sold for more than its depreciated book value there is a tax bill, and vice versa
- E.g., in the data center example, suppose it could be sold for \$500 million at $t = 5$. It has been depreciated to \$200 million, so the company pays taxes on the gain,

$$(500 - 200) \times 21\% = \$63 \text{ million}$$

- The net (after-tax) proceeds are $500 - 63 = \$437$ million, and now the NPV is positive ($+\$30.48$ million)

# Example, the Data Center with a Higher Terminal Cash Flow

- Solution ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| Revenues − Cash expenses | | 200 | 200 | 200 | 200 | 200 |
| Depreciation | | 160 | 160 | 160 | 160 | 160 |
| EBIT | | 40 | 40 | 40 | 40 | 40 |
| Net income | | 31.6 | 31.6 | 31.6 | 31.6 | 31.6 |
| + Depreciation | | 160 | 160 | 160 | 160 | 160 |
| Operating cash flows | | 191.6 | 191.6 | 191.6 | 191.6 | 191.6 |
| Capital investment | −1,000 | | | | | |
| Sale of the data center (after tax) | | | | | | 437 |
| Working capital | −100 | | | | | 100 |
| **Total cash flows** | −1,100 | 191.6 | 191.6 | 191.6 | 191.6 | 728.6 |
| **NPV** | 30.48 | | | | | |
| **IRR** | 8.8% | | | | | |

# Rule f, Terminal Cash Flows Can Be Negative

- Terminal cash flows can also be negative, e.g., if we have to pay to shut down the project



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- Terminal cash flows can also be negative, e.g., if we have to pay to shut down the project
- E.g., oil wells require "capping" when they no longer produce. Capping is expensive, so oil well projects are worth more if you do not cap them (although this is illegal!)
- But apparently many companies still do this, Bloomberg, "It Pays to Not Pay Your Debts"



# Practice Problem

- A firm plans a new manufacturing plant on a site it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. The market value of the site
b. The market value of the existing buildings
c. Demolition costs and site clearance
d. The cost of a new access road put in last year
e. Lost cash flows on other projects due to executive time spent on the new facility
f. Future depreciation of the new plant
g. The reduction in the firm's tax bill resulting from depreciation of the new plant
h. The initial investment in inventories of raw materials
i. Money already spent on engineering design of the new plant

# Practice Problem

a, b. The site and buildings could have been sold or put to another use. Their values are opportunity costs, treated as incremental cash outflows

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i. The expenditure on design is a sunk cost and not an incremental cash flow

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- Only incremental cash flows count, would the cash flow exist without the project?
- Count indirect effects, opportunity costs, working capital, and terminal cash flows (with the tax on any gain over book value), ignore sunk costs and allocated overhead

## Module 7, Part 4

# Full-Scale Capital Budgeting Problems

# Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - It costs 1,000 and lasts 5 years, depreciated straight-line to a salvage value of 200, and sold at $t = 5$ for 200

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- This time, build the table step by step

# Example, the Data Center Again

- Solution ($ millions), the operating cash flows first, then working capital, then the capital investment and the sale, then the totals

| Year | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|

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| Depreciation | | 160 | 160 | 160 | 160 | 160 |
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| Working capital | −100 | | | | | 100 |

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| Operating cash flows | | 191.6 | 191.6 | 191.6 | 191.6 | 191.6 |
| Working capital | −100 | | | | | 100 |
| Capital investment | −1,000 | | | | | |
| Sale of the data center | | | | | | 200 |
| **Total cash flows** | −1,100 | 191.6 | 191.6 | 191.6 | 191.6 | 491.6 |
| **NPV** | −130.82 | | | | | |

- NPV = −$130.82 million, reject the project, the same answer as in Part 2

# How to Attack a Full-Scale Problem

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3. Add the project's capital investment at the start and the terminal cash flows at the end (after tax)
4. Total the cash flows year by year, discount at the discount rate, and apply the NPV rule

# Example 1, Growing Sales and a Taxed Sale of Equipment

- A company makes bottles and just developed a new bottle design
- The initial investment in machinery is $6 million. The equipment is depreciated straight-line over 6 years to zero, but will actually be sold in 6 years for $620,000 (so there are tax consequences)

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- Sales are projected at 1 million bottles in the first year, growing at 5% a year over the six years. Production costs are \$2.00 per bottle, and each bottle sells for \$4.00
- The new design is phased out after 6 years. The tax rate is 21%, and the required rate of return is 12%
- What is the project's NPV?

# Example 1, Growing Sales and a Taxed Sale of Equipment

- Operating cash flows ($ thousands),

| Year | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Sales (thousand bottles) | 1,000 | 1,050 | 1,103 | 1,158 | 1,216 | 1,276 |
| Sales ($) | 4,000 | 4,200 | 4,410 | 4,631 | 4,862 | 5,105 |
| Cost ($) | 2,000 | 2,100 | 2,205 | 2,315 | 2,431 | 2,553 |
| Depreciation | 1,000 | 1,000 | 1,000 | 1,000 | 1,000 | 1,000 |
| EBIT | 1,000 | 1,100 | 1,205 | 1,315 | 1,431 | 1,553 |
| Taxes | 210 | 231 | 253 | 276 | 301 | 326 |
| Net income | 790 | 869 | 952 | 1,039 | 1,130 | 1,227 |
| + Depreciation | 1,000 | 1,000 | 1,000 | 1,000 | 1,000 | 1,000 |
| **Operating cash flows** | 1,790 | 1,869 | 1,952 | 2,039 | 2,130 | 2,227 |

# Example 1, Growing Sales and a Taxed Sale of Equipment

- Terminal cash flow from the sale of the equipment ($ thousands), the book value is zero so the whole sale price is taxed,

| Terminal cash flows | $ |
|---|---|
| Sales price | 620 |
| Tax | 130 |
| Net income from sale | 490 |

# Example 1, Growing Sales and a Taxed Sale of Equipment

- Total cash flows ($ thousands),

| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| Capital | −6,000 | | | | | | 490 |
| Working capital | −250 | | | | | | 250 |
| Operating cash flows | | 1,790 | 1,869 | 1,952 | 2,039 | 2,130 | 2,227 |
| **Total cash flows** | −6,250 | 1,790 | 1,869 | 1,952 | 2,039 | 2,130 | 2,966 |
| PV | −6,250 | 1,598 | 1,490 | 1,389 | 1,296 | 1,209 | 1,503 |
| **NPV** | 2,235 | | | | | | |

# Example 1, Growing Sales and a Taxed Sale of Equipment

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| PV | −6,250 | 1,598 | 1,490 | 1,389 | 1,296 | 1,209 | 1,503 |
| **NPV** | 2,235 | | | | | | |

- The Excel solution is in the "Bottles" tab of Module 7 Excel Example.xlsx

# Practice Problem

- A firm has developed a new mousetrap. It can go into production for an initial investment in equipment of \$6.3 million. The equipment will be depreciated straight-line over 6 years, but in fact can be sold after 6 years for \$695,000
- Working capital at each date must be maintained at 10% of next year's forecast sales. Production costs are \$1.80 per trap and the traps sell for \$7 each. The project ends in 6 years when the trap becomes obsolete. The tax rate is 40%, and the required rate of return is 11%

| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 | After |
|---|---|---|---|---|---|---|---|---|
| Sales (million traps) | 0 | 0.6 | 0.8 | 0.9 | 0.9 | 0.5 | 0.2 | 0 |

# Practice Problem

- Suppose the firm can cut its working capital requirements in half with better inventory control. By how much does this increase the project NPV?

# Practice Problem

- Suppose the firm can cut its working capital requirements in half with better inventory control. By how much does this increase the project NPV?
- Solution ($ millions), working capital at each date is 10% of next year's sales,

| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| Sales units | | 0.600 | 0.800 | 0.900 | 0.900 | 0.500 | 0.200 |
| Revenue | 0.000 | 4.200 | 5.600 | 6.300 | 6.300 | 3.500 | 1.400 |
| NWC | 0.420 | 0.560 | 0.630 | 0.630 | 0.350 | 0.140 | 0.000 |
| Cash flow from NWC | −0.420 | −0.140 | −0.070 | 0.000 | 0.280 | 0.210 | 0.140 |

# Practice Problem

- With working capital requirements halved ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| New NWC | 0.210 | 0.280 | 0.315 | 0.315 | 0.175 | 0.070 | 0.000 |
| New cash flow from NWC | −0.210 | −0.070 | −0.035 | 0.000 | 0.140 | 0.105 | 0.070 |
| Change in cash flow | 0.210 | 0.070 | 0.035 | 0.000 | −0.140 | −0.105 | −0.070 |

# Practice Problem

- With working capital requirements halved ($ millions),

| Year | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| New NWC | 0.210 | 0.280 | 0.315 | 0.315 | 0.175 | 0.070 | 0.000 |
| New cash flow from NWC | −0.210 | −0.070 | −0.035 | 0.000 | 0.140 | 0.105 | 0.070 |
| Change in cash flow | 0.210 | 0.070 | 0.035 | 0.000 | −0.140 | −0.105 | −0.070 |

- Discounting the changes in cash flow at 11%, the change in NPV is $0.1095 million

# Practice Problem

- A firm may upgrade its modem pool. It last upgraded 2 years ago, when it spent $140 million on equipment with an assumed life of 5 years and an assumed salvage value of $20 million for tax purposes. The firm uses straight-line depreciation. The old equipment can be sold today for $100 million
- A new modem pool can be set up today for $210 million. It has a 3-year life and is depreciated to zero straight-line. The new equipment increases sales by $28 million per year and decreases operating costs by $14 million per year. At the end of 3 years it is worthless. The tax rate is 30% and the discount rate is 9%

# Practice Problem

a. What is the net cash flow at time 0 if the old equipment is replaced?
b. What are the incremental cash flows in years 1, 2, and 3?
c. What is the NPV of the replacement project?
d. What is the IRR of the replacement project?

# Practice Problem

- a. Annual depreciation is $(\$140 - \$20)/5 = \$24$. Book value at the time of sale is $\$140 - 2 \times \$24 = \$92$

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- a. Annual depreciation is $(\$140 - \$20)/5 = \$24$. Book value at the time of sale is $\$140 - 2 \times \$24 = \$92$

Sales price = $100, so net-of-tax proceeds from the sale are:
$100 − [0.30 × ($100 − $92)] = $97.60

Therefore, the net cash outlay at time 0 is $210 − $97.60 = −$112.40.

**b.** The project saves $14 in operating costs and increases sales by $28. The depreciation expense for the new machine would be $70 per year.

Therefore, including the depreciation tax shield, operating cash flow increases by:
($28 + $14) × (1 − 0.30) + ($70 × 0.30) = $50.40 per year

**c.** NPV = −$112.40 + $50.40 × ((1 ÷ 0.09) − {1 ÷ [0.09(1 + 0.09)$^3$]})
= $15.18

**d.** To find the internal rate of return, find the interest rate "r" that makes the Present Value of the inflows equal and opposite to the initial outflow. To do this set the PV of the three year annuity to $112.40 and solve for the discount rate (r):

$112.40 = $50.40 × [1 ÷ r − 1 ÷ r × (1 + r)$^3$]
r = 16.43%

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- Use cash flows, not accounting profits, treat financing separately, and remember the depreciation tax shield
- Only incremental cash flows count, include indirect effects, opportunity costs, working capital, and terminal cash flows, ignore sunk costs
- Lay the years out in a table, total the cash flows, discount, and apply the NPV rule

# Thank you!

Please let me know if you have any questions.